

The Era of Large Models and Expanding Horizons

– Applications and Challenges –
Elliott Wu

GPT-4



SORA



SAM 3



SAM 3D



Genie



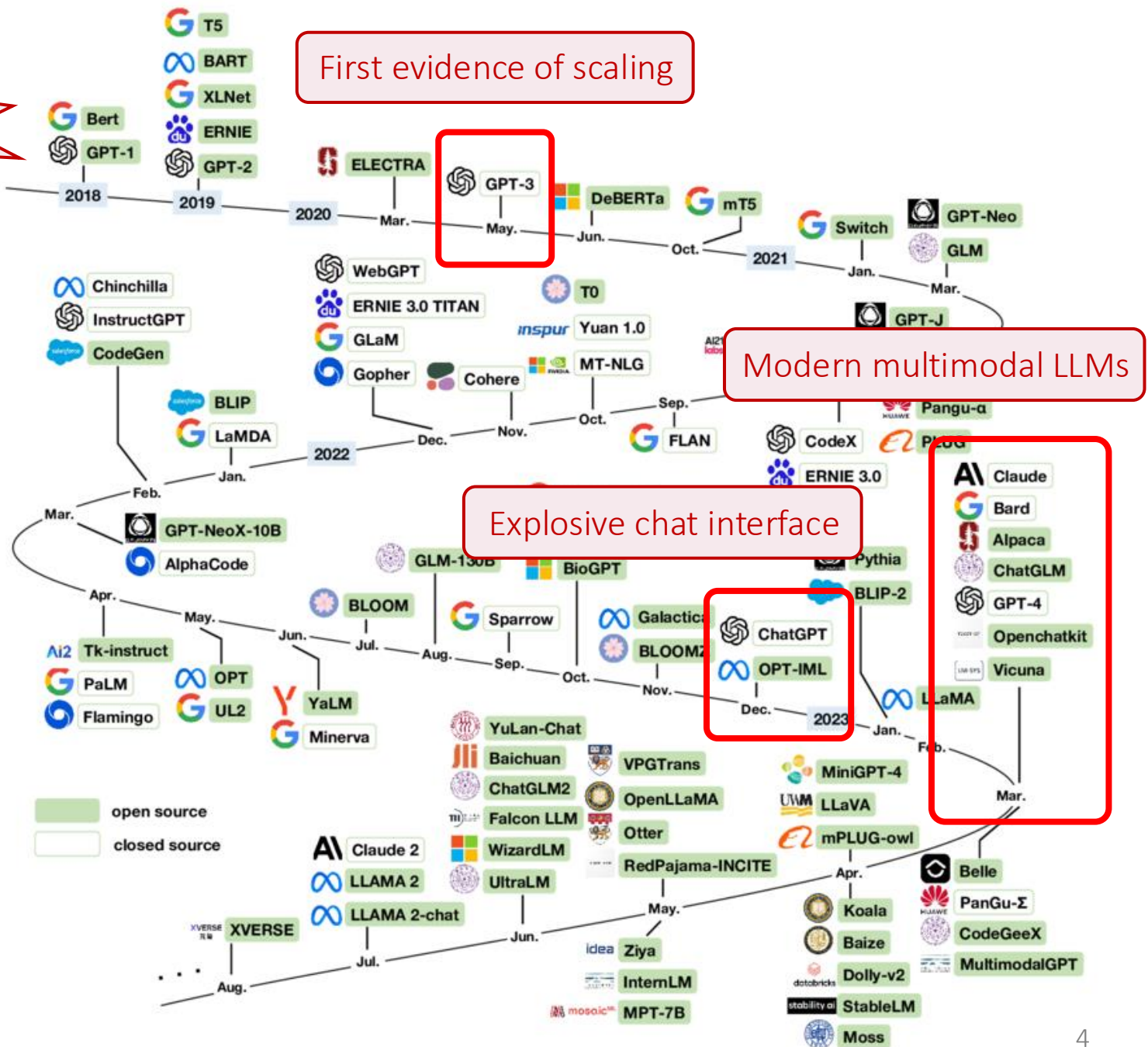
Applications and Challenges – Outline

- Part 1 – The Era of Large Models
- Part 2 – Expanding Horizons

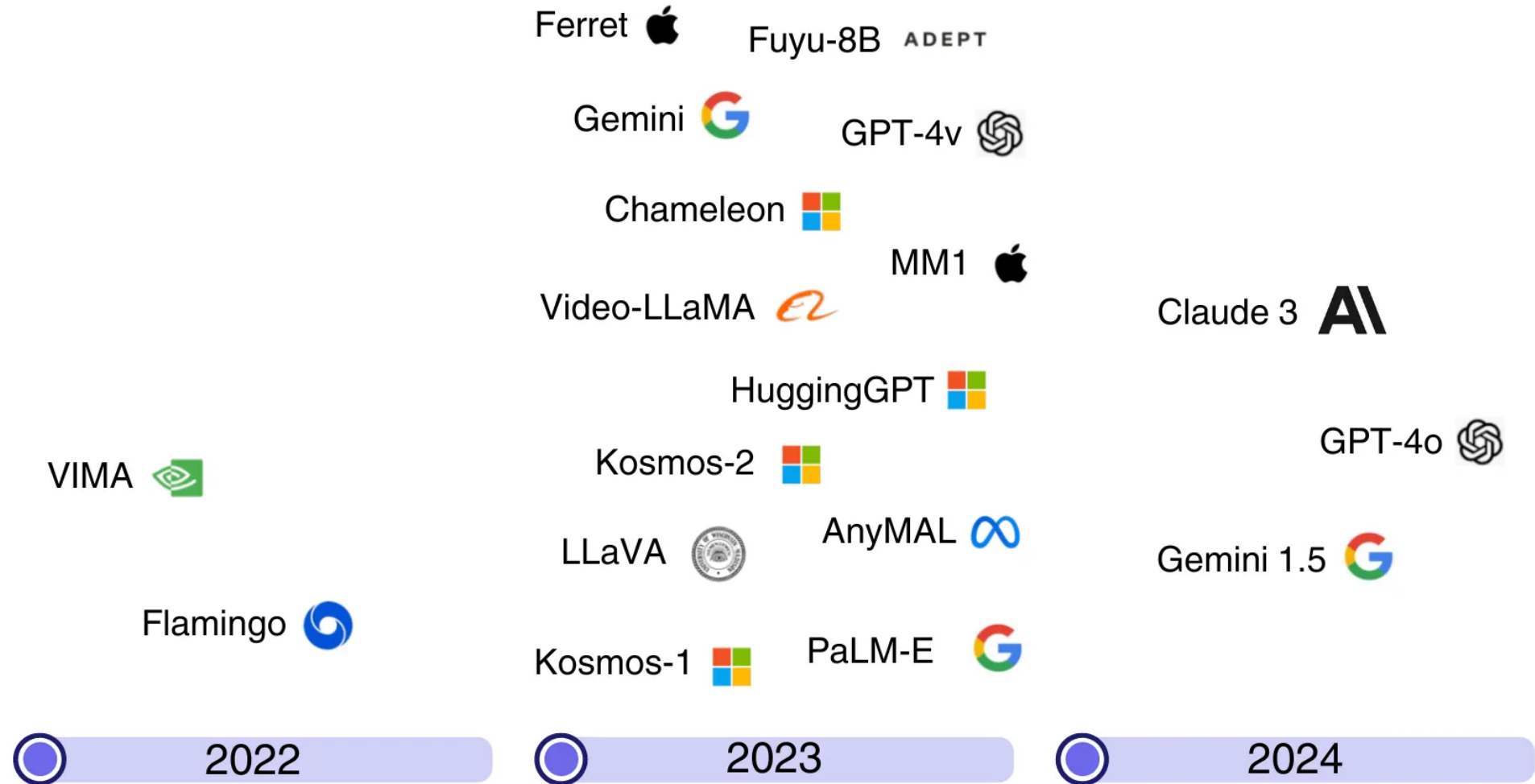
Part 1 – The Era of Large Models

A Brief History of LLMs (2018-2023)

- GPT-1 (2018): 117M params
- GPT-2 (2019): 1.5B params
- GPT-3 (2020): 175B params
- GPT-4 (2023): >1T params

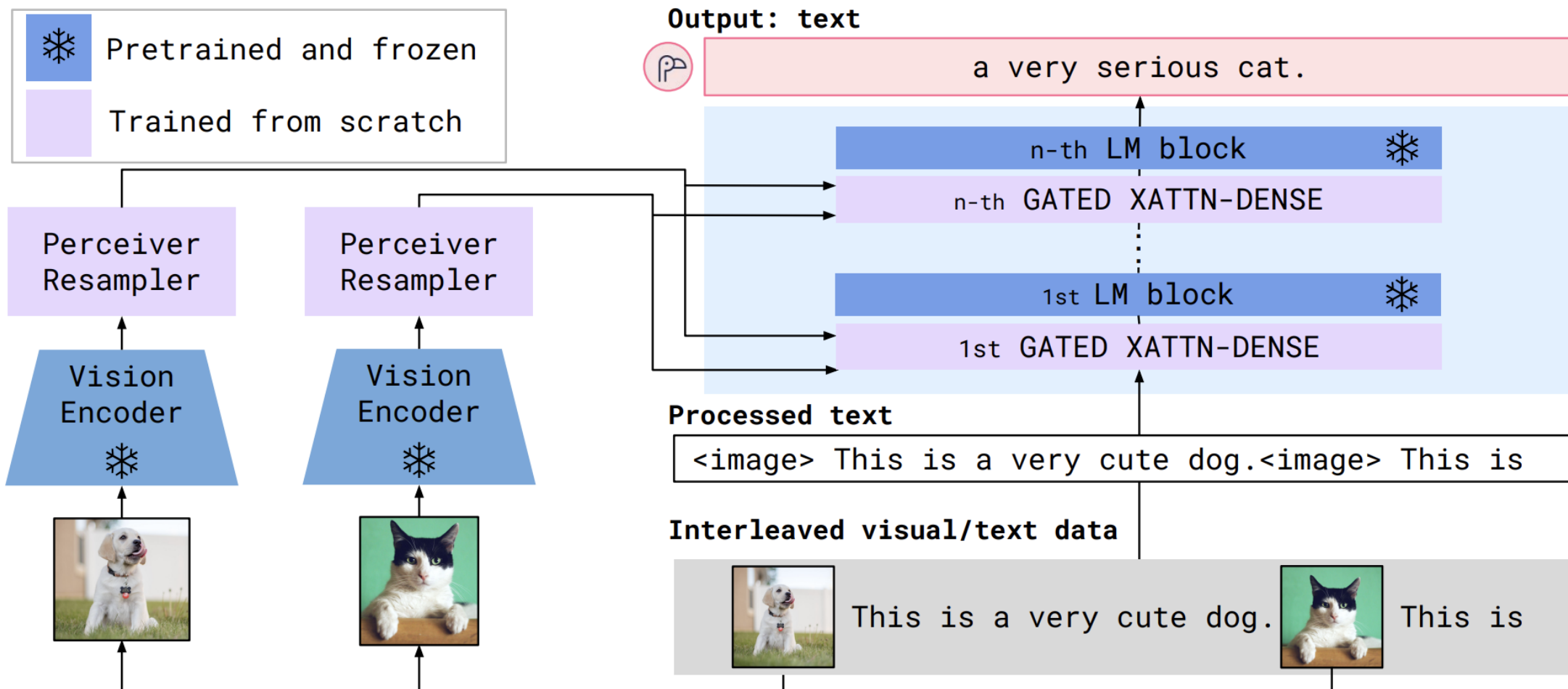


Multimodal Language Models (2022-2024)

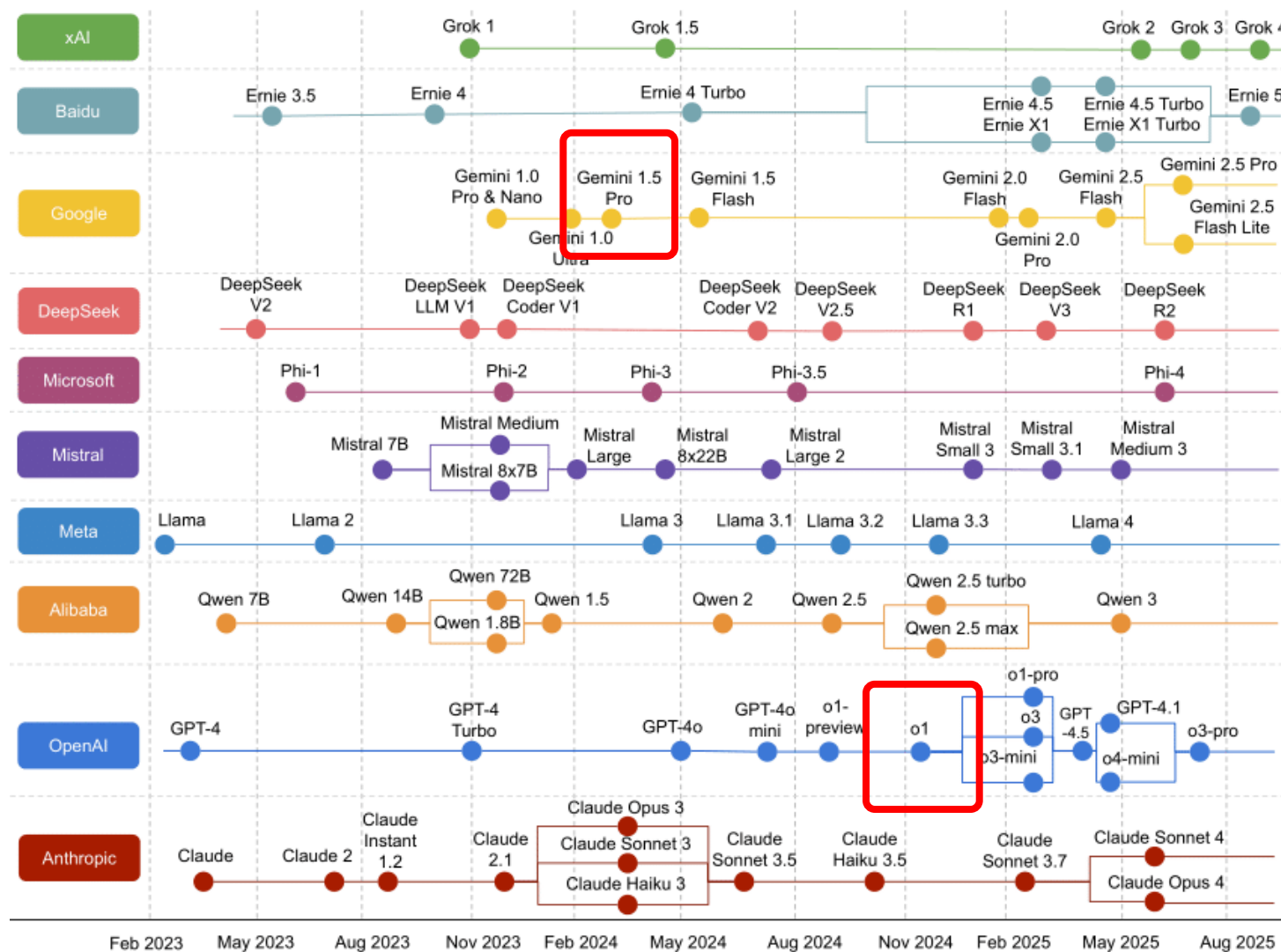


https://medium.com/@tenyks_blogger/multimodal-large-language-models-mlms-transforming-computer-vision-76d3c5dd267f

Flamingo: a Visual Language Model for Few-Shot Learning



Reasoning Models (2024-2025)



1M token context window
(hour-long video)

Chain-of-Thought (CoT)
reasoning, solving 83%
of IMO questions

As of 5 March 2026

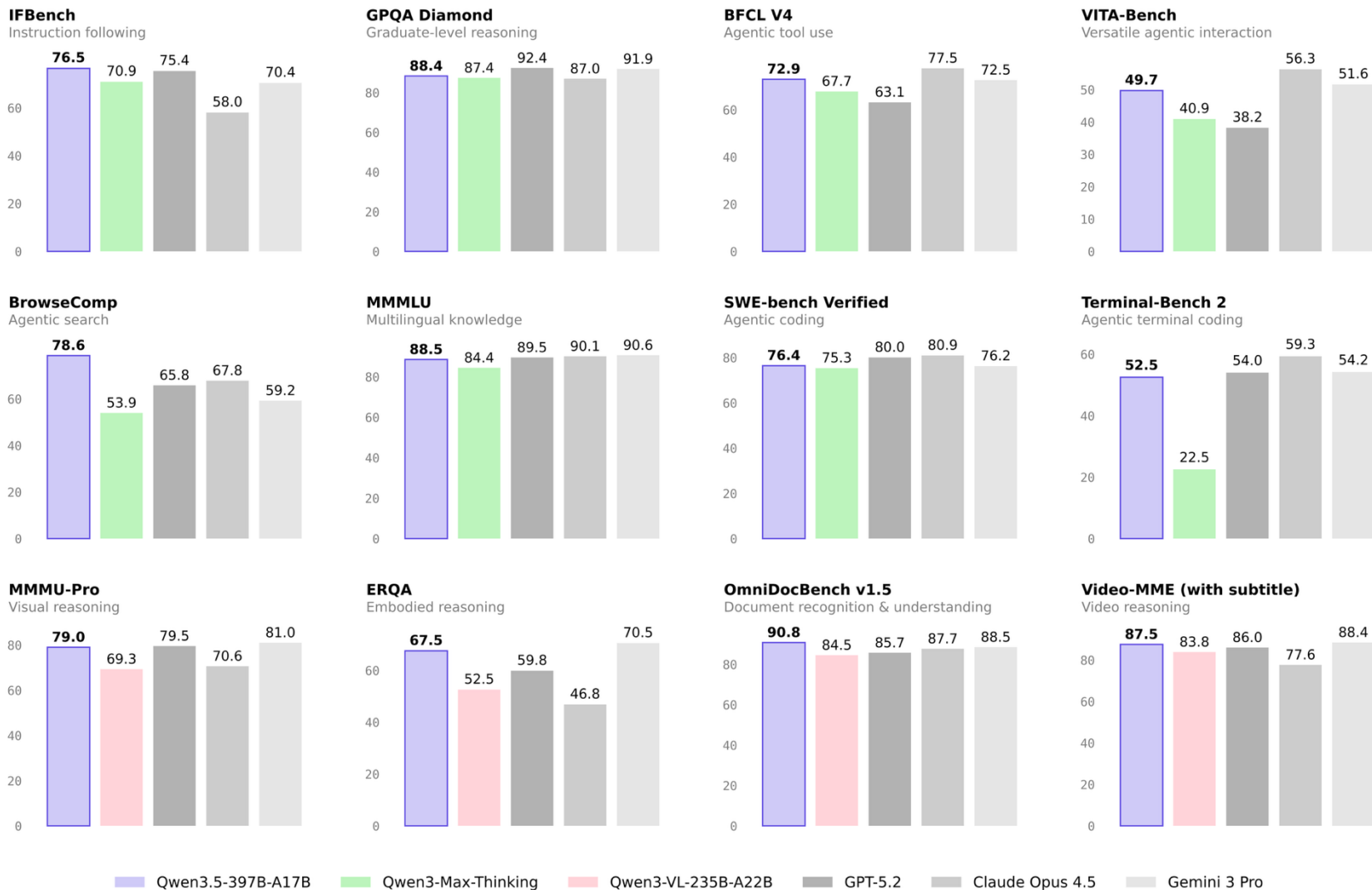
Closed-source (proprietary)

- Gemini 3.1 Pro (Google)
- GPT 5.4 (OpenAI)
- Claude 4.6 (Anthropic)
- Grok 4.1 (xAI)
- Seed 2.0 (ByteDance)
- ...

Open-weight

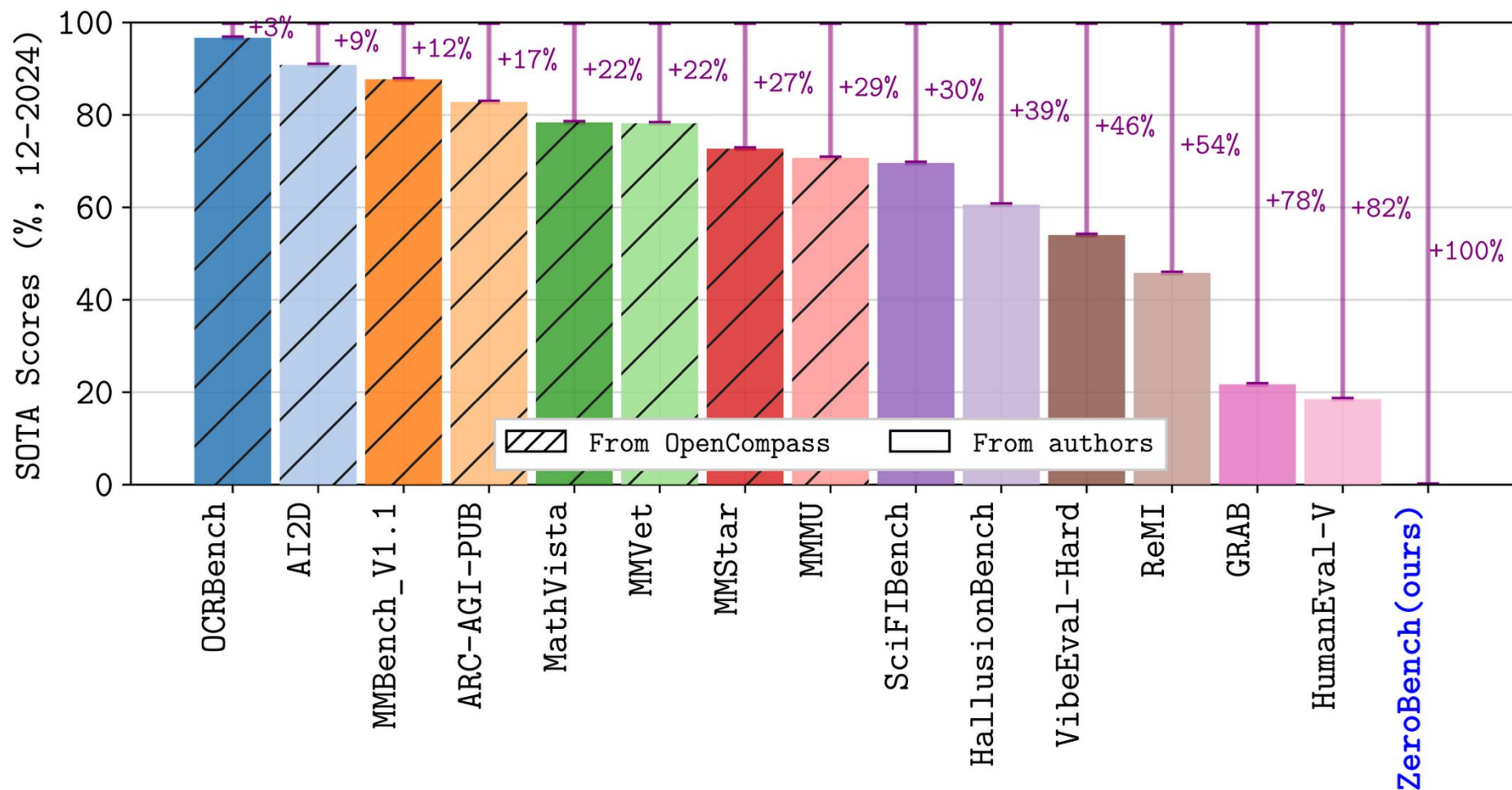
- Qwen 3.5 (Alibaba)
- Kimi K2.5 (Moonshot AI)
- DeepSeek V3.2 (DeepSeek)
- GLM 5 (Zhipu)
- Gemma 3 (Google)
- Mistral 3 (Mistral AI)
- Llama 4 (Meta)
- GPT-OSS-120B (OpenAI)
- ...

Evaluation Benchmarks



ZeroBench

An Impossible* Visual Benchmark for Contemporary Large Multimodal Models



ZeroBench

An Impossible* Visual Benchmark for Contemporary Large Multimodal Models

ZeroBench					
Models	Main questions (100)			Subquestions (334)	
	k/k [%] (n)	pass@k [%] (n)		pass@k [%] (SE_{CLT})	Num. correct
	k=5	k=1	k=5	k=1	k=1
Reasoning LMMs					
o1 pro ^{◇,§}	0.0 (0)	0.0 (0)	-	22.40 (2.48)	75
o1 [◇]	0.0 (0)	0.0 (0)	0.0 (0)	20.21 (2.33)	68
Gemini 2 Flash Thinking	0.0 (0)	0.0 (0)	5.0 (5)	20.51 (2.60)	69
QVQ	0.0 (0)	0.0 (0)	4.0 (4)	20.47 (2.38)	70
Proprietary LMMs					
GPT-4o	0.0 (0)	0.0 (0)	0.0 (0)	19.60 (2.37)	67
GPT-4o mini	0.0 (0)	0.0 (0)	2.0 (2)	16.58 (2.41)	54
Gemini 2 Flash	0.0 (0)	0.0 (0)	3.0 (3)	23.24 (2.85)	75
Gemini 1.5 Pro	0.0 (0)	0.0 (0)	2.0 (2)	20.88 (2.47)	74
Gemini 1.5 Flash	0.0 (0)	0.0 (0)	2.0 (2)	17.87 (2.41)	63
Gemini 1 Pro Vision	0.0 (0)	0.0 (0)	2.0 (2)	12.36 (2.08)	46
Claude 3.5 Sonnet v2	0.0 (0)	0.0 (0)	2.0 (2)	25.50 (2.67)	82
Claude 3.5 Sonnet	0.0 (0)	0.0 (0)	1.0 (1)	20.71 (2.48)	72
Claude 3 Opus	0.0 (0)	0.0 (0)	0.0 (0)	15.10 (2.16)	45
Claude 3 Sonnet	0.0 (0)	0.0 (0)	1.0 (1)	16.08 (2.25)	49
Claude 3 Haiku	0.0 (0)	0.0 (0)	0.0 (0)	12.27 (2.05)	41
Reka Edge	0.0 (0)	0.0 (0)	0.0 (0)	3.74 (0.96)	13
Open-weight LMMs					
Llama 3.2 90B	0.0 (0)	0.0 (0)	0.0 (0)	13.26 (1.92)	48
Qwen2-VL-72B-Instruct	0.0 (0)	0.0 (0)	2.0 (2)	13.00 (2.32)	43
NVLM-D-72B	0.0 (0)	0.0 (0)	1.0 (1)	14.91 (2.36)	51
Pixtral-Large	0.0 (0)	0.0 (0)	3.0 (3)	18.68 (2.26)	62

ZeroBench

An Impossible* Visual Benchmark for Contemporary Large Multimodal Models

Model	Main questions (100)			Subquestions (334)	
	pass@1	pass@5	pass^5	pass@1	n correct
Gemini 3.1 Pro 🔥	-	19.0	7.0	-	-
Gemini 3 Pro 🔥	-	19.0	5.0	-	-
GPT-5.2 (medium reasoning) 🔥	-	17.0	6.0	-	-
Gemini 3 Flash 🔥	-	13.0	2.0	-	-
Claude Opus 4.5 🔥	-	10.0	1.0	-	-
o4-mini 🔥	2.0	10.0	0.0	29.05	89
GPT-5-mini (high reasoning) 🔥	4.0	9.0	3.0	27.79	95
Claude Opus 4.1 (thinking) 🔥	5.0	8.0	1.0	24.38	80
GPT-5 (medium reasoning) 🔥	1.0	7.0	0.0	26.19	93
GPT 4.5 🔥	1.0	7.0	0.0	27.01	95
Claude Sonnet 4 (thinking) 🔥	3.0	6.0	0.0	24.60	80
o3 🔥	3.0	6.0	0.0	25.51	85
GPT-5.1 (medium reasoning) 🔥	-	5.0	0.0	-	-
Claude Opus 4 (thinking) 🔥	4.0	5.0	1.0	25.06	81
Claude Sonnet 4 🔥	2.0	5.0	1.0	23.01	77
Gemini 2.5 Pro 🔥	3.0	5.0	1.0	26.04	90
Gemini 2 Flash Thinking	0.0	5.0	0.0	20.51	69
Claude Opus 4.1 🔥	1.0	4.0	1.0	25.30	81
Grok 4 🔥	1.0	4.0	0.0	21.58	71

ZeroBench

An Impossible* Visual Benchmark for Contemporary Large Multimodal Models



Question:

In the image there is a stationary pot (containing different pens and a letter opener) and other things. Consider: (1) The typical number of legs of the species visible in the picture (2) The total number of pens (3) The total number of pen nibs exposed and visible. What is the value of the product of (1), (2) and (3)?

ZeroBench

An Impossible* Visual Benchmark for Contemporary Large Multimodal Models

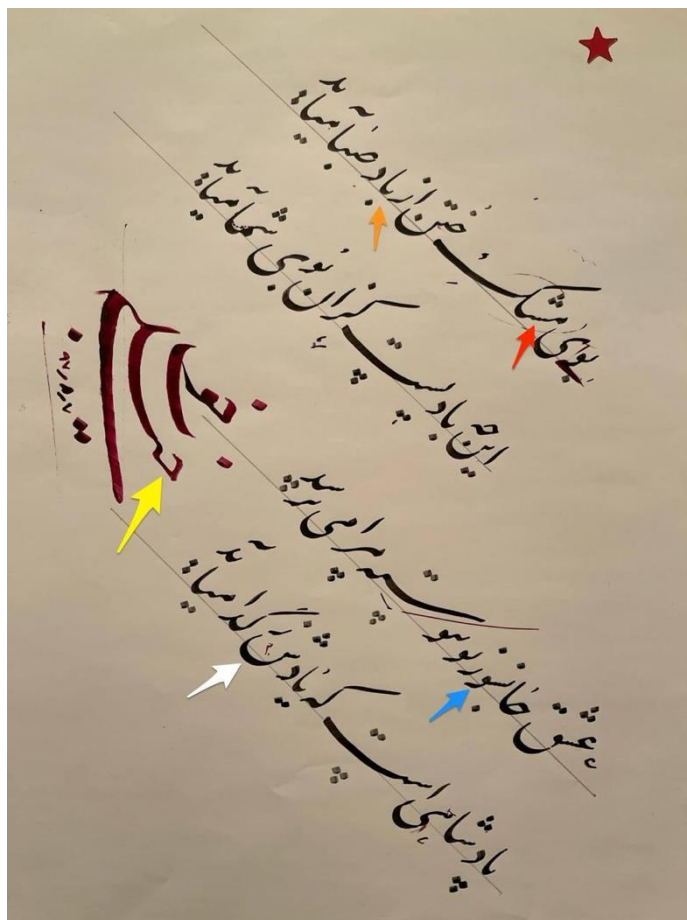


Question:

Read the note pointed at by the pink arrow. Then, read the note pointed at by the green arrow. Return the two notes in that order without a space or comma.

ZeroBench

An Impossible* Visual Benchmark for Contemporary Large Multimodal Models



Question:

The image shows calligraphic writing in Farsi, with some letters annotated with arrows. If we put these letters together, what would the message be when translated into English? Use the colors in the following order: red, orange, yellow, blue, and white.

Qwen 3.5 Small Models (3 March 2026)

 **Qwen/Qwen3.5-0.8B** NEW

 Image-Text-to-Text  Safetensors  PyTorch ...

 Qwen |  2026.03.03 |  15.2k |  10

 **Qwen/Qwen3.5-2B** NEW

 Image-Text-to-Text  Safetensors  PyTorch ...

 Qwen |  2026.03.03 |  4.2k |  3

 **Qwen/Qwen3.5-4B** NEW

 Image-Text-to-Text  Safetensors  PyTorch ...

 Qwen |  2026.03.03 |  19.2k |  11

 **Qwen/Qwen3.5-9B** NEW

 Image-Text-to-Text  Safetensors  PyTorch ...

 Qwen |  2026.03.03 |  19.2k |  15

Qwen3.5-0.8B WebGPU demo



Qwen just released their 0.8B model.
So naturally, I had to try running it on my 7-year-old Samsung S10E.

https://www.reddit.com/r/LocalLLaMA/comments/1rizodv/running_qwen_35_08b_locally_in_the_browser_on/

Qwen 3.5 Small Models

 **Qwen/Qwen3.5-0.8B** NEW

 Image-Text-to-Text  Safetensors  PyTorch ...

 Qwen |  2026.03.03 |  15.2k |  10

 **Qwen/Qwen3.5-2B** NEW

 Image-Text-to-Text  Safetensors  PyTorch ...

 Qwen |  2026.03.03 |  4.2k |  3

 **Qwen/Qwen3.5-4B** NEW

 Image-Text-to-Text  Safetensors  PyTorch ...

 Qwen |  2026.03.03 |  19.2k |  11

 **Qwen/Qwen3.5-9B** NEW

 Image-Text-to-Text  Safetensors  PyTorch ...

 Qwen |  2026.03.03 |  19.2k |  15

Fine-tune Qwen3.5

Train Qwen3.5 locally on just 5GB VRAM.

You can now fine-tune [Qwen3.5](#) model family (0.8B, 2B, 4B, 9B, 27B, 35B-A3B, 122B-A10B) with [Unsloth](#). Support includes [vision](#) and text. **Qwen3.5-35B-A3B** - bf16 LoRA works on **74GB VRAM**.

- Unsloth makes Qwen3.5 train **1.5x faster** and uses **50% less VRAM** than FA2 setups.
- Qwen3.5 BF16 LoRA VRAM requirements: **0.8B**: 3GB • **2B**: 5GB • **4B**: 10GB • **27B**: 56GB
- Fine-tune **0.8B, 2B, 4B** and **9B** via our **free Google Colab notebooks**:

[Qwen3.5-0.8B](#)

[Qwen3.5-2B](#)

[Qwen3.5-4B](#)



- If you want to **preserve reasoning** ability, you can mix reasoning-style examples with direct answers (keep a minimum of 75% reasoning). Otherwise you can emit it fully.
- After fine-tuning, you can export to [GGUF](#) (for llama.cpp/Ollama/LM Studio/etc.) or [vLLM](#)
- [Reinforcement Learning](#) (RL) for Qwen3.5 [VLM RL](#) also works via Unsloth inference.
- We have **A100** Colab notebooks for [Qwen3.5-27B](#) and [Qwen3.5-35B-A3B](#).

If you're on an older version (or fine-tuning locally), update first:

```
pip install --upgrade --force-reinstall --no-cache-dir unsloth unsloth_zoo
```

⚠ If training seems **slower than usual**, it's because Qwen3.5 use custom Mamba Triton kernels. Compiling those kernels can take longer than normal, especially on T4 GPUs.

Below is a minimal SFT recipe (works for "text-only" fine-tuning). See also [vision fine-tuning](#) section.

```
from unsloth import FastLanguageModel
import torch
from datasets import load_dataset
from trl import SFTTrainer, SFTConfig

max_seq_length = 2048 # start small; scale up after it works

# Example dataset (replace with yours). Needs a "text" column.
url = "https://huggingface.co/datasets/laion/OIG/resolve/main/unified_chip2.jsonl"
dataset = load_dataset("json", data_files={"train": url}, split="train")
```

In-context Learning

Language Models are Few-Shot Learners

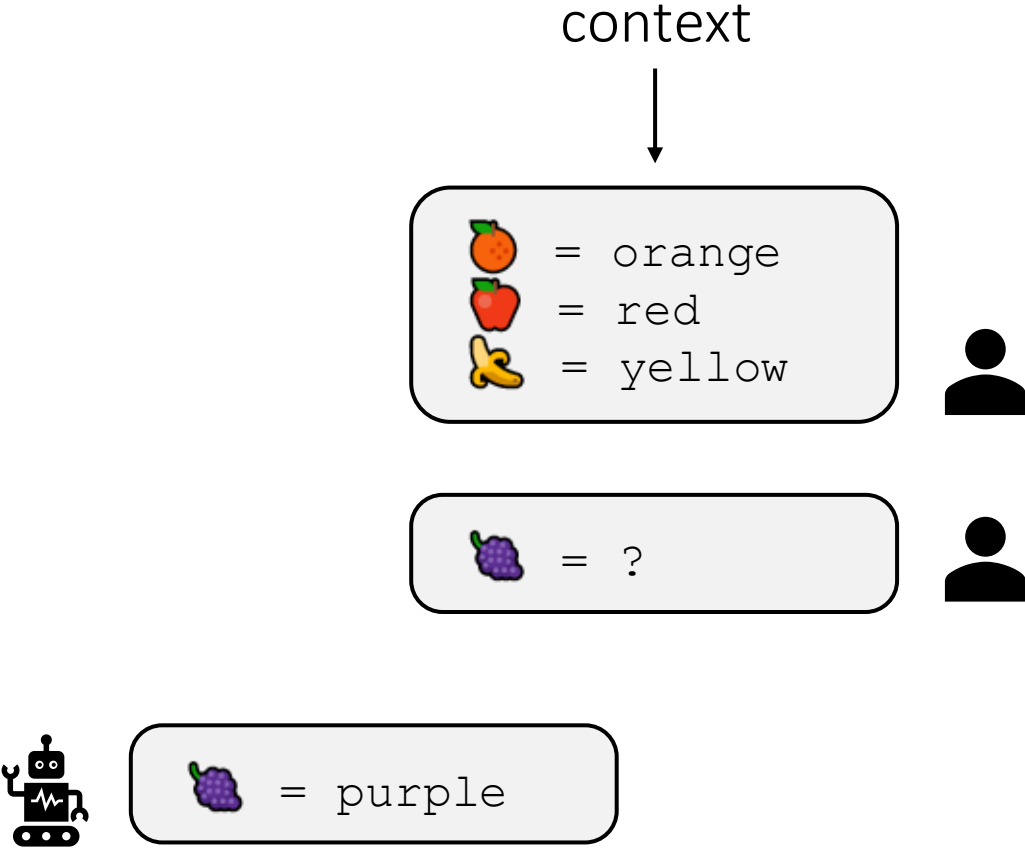
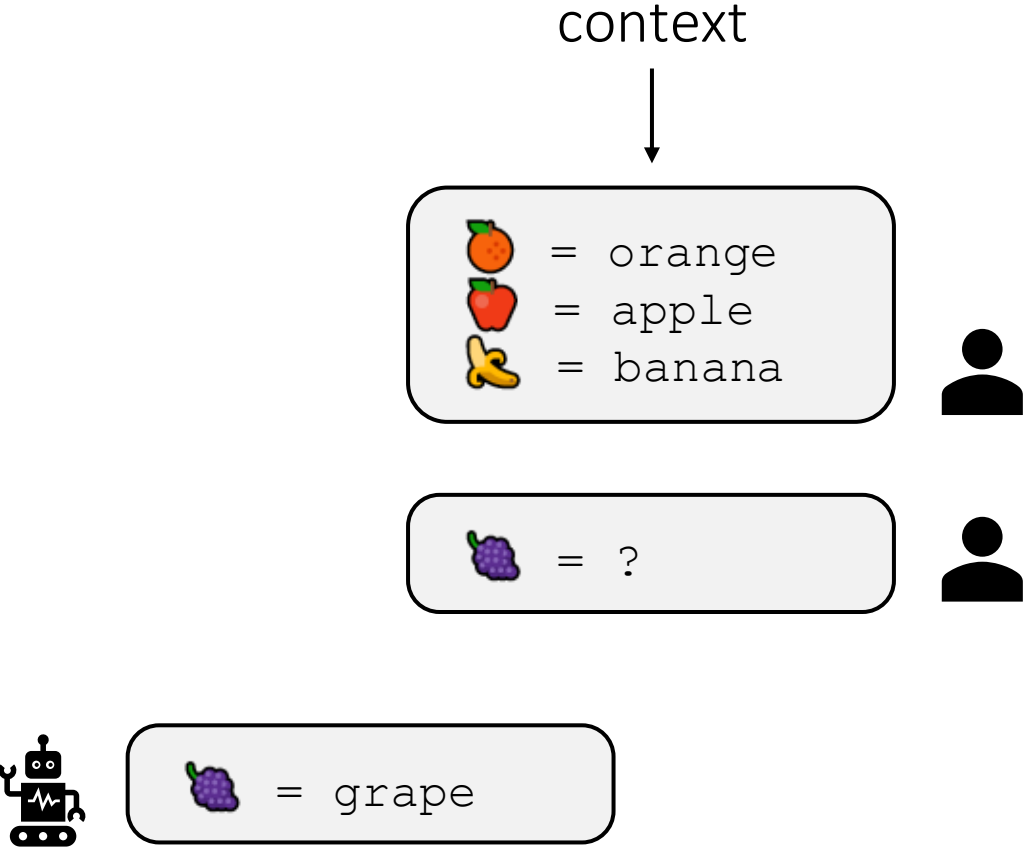
OpenAI

GPT-3 Paper

Abstract

Recent work has demonstrated substantial gains on many NLP tasks and benchmarks by pre-training on a large corpus of text followed by fine-tuning on a specific task. While typically task-agnostic in architecture, this method still requires task-specific fine-tuning datasets of thousands or tens of thousands of examples. By contrast, humans can generally perform a new language task from only a few examples or from simple instructions – something which current NLP systems still largely struggle to do. Here we show that scaling up language models greatly improves task-agnostic, few-shot performance, sometimes even reaching competitiveness with prior state-of-the-art fine-tuning approaches. Specifically, we train GPT-3, an autoregressive language model with 175 billion parameters, 10x more than any previous non-sparse language model, and test its performance in the few-shot setting. For all tasks, GPT-3 is applied without any gradient updates or fine-tuning, with tasks and few-shot demonstrations specified purely via text interaction with the model. GPT-3 achieves strong performance on many NLP datasets, including translation, question-answering, and cloze tasks, as well as several tasks that require on-the-fly reasoning or domain adaptation, such as unscrambling words, using a novel word in a sentence, or performing 3-digit arithmetic. At the same time, we also identify some datasets where GPT-3's few-shot learning still struggles, as well as some datasets where GPT-3 faces methodological issues related to training on large web corpora. Finally, we find that GPT-3 can generate samples of news articles which human evaluators have difficulty distinguishing from articles written by humans. We discuss broader societal impacts of this finding and of GPT-3 in general.

In-context Learning



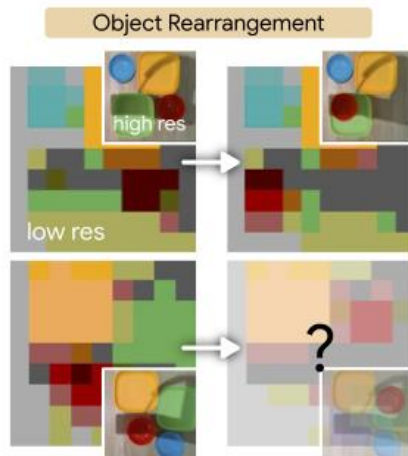
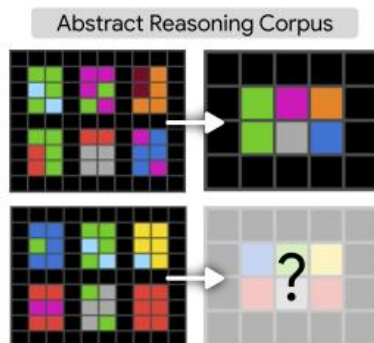
Large Language Models as General Pattern Machines

Suvir Mirchandani Fei Xia Pete Florence Brian Ichter Danny Driess Montserrat Gonzalez Arenas
Kanishka Rao Dorsa Sadigh Andy Zeng



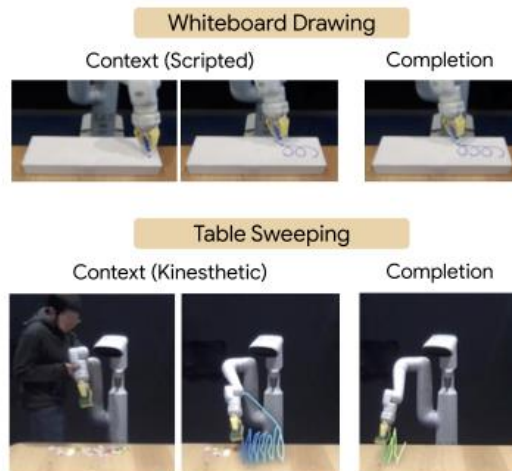
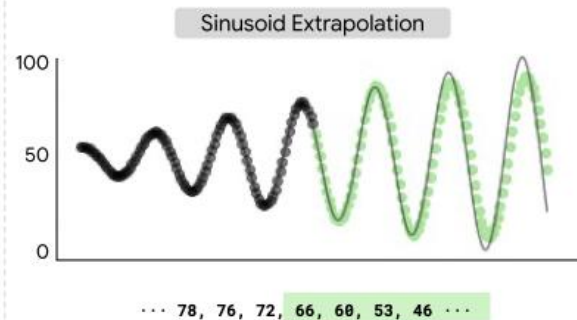
Sequence Transformation

Pattern transformations (symbolic)



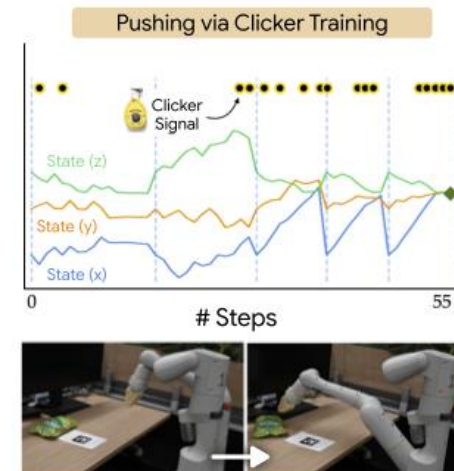
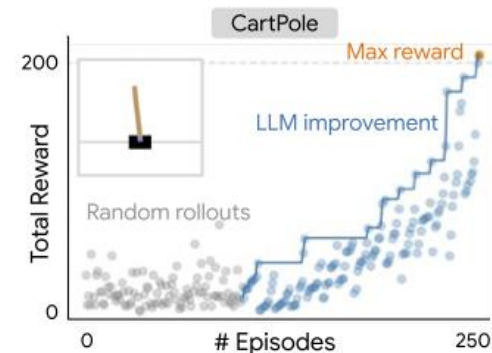
Sequence Completion

Simple function classes (numeric)



Sequence Improvement

Online policies (numeric & symbolic)



An Explanation of In-context Learning as Implicit Bayesian Inference

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Abstract

Large language models (LMs) such as GPT-3 have the surprising ability to do in-context learning, where the model learns to do a downstream task simply by conditioning on a prompt consisting of input-output examples. The LM learns from these examples *without being explicitly pretrained to learn*. Thus, it is unclear what enables in-context learning. In this paper, we study how in-context learning can emerge when pretraining documents have long-range coherence. Here, the LM must infer a latent document-level concept to generate coherent next tokens during pretraining. At test time, in-context learning occurs when the LM also infers a shared latent concept between examples in a prompt. We prove when this occurs despite a distribution mismatch between prompts and pretraining data in a setting where the pretraining distribution is a mixture of HMMs. In contrast to messy large-scale datasets used to train LMs capable of in-context learning, we generate a small-scale synthetic dataset (GINC) where Transformers and LSTMs both exhibit in-context learning¹. Beyond the theory, experiments on GINC exhibit large-scale real-world phenomena including improved in-context performance with model scaling (despite the same pretraining loss), sensitivity to example order, and instances where zero-shot is better than few-shot in-context learning.

Agentic Context Engineering: Evolving Contexts for Self-Improving Language Models

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Vamsidhar Kamanuru² Jay Rainton² Chen Wu² Mengmeng Ji² Hanchen Li³
Urmish Thakker² James Zou¹ Kunle Olukotun¹

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🌐 <https://github.com/ace-agent/ace> 🌐 <https://ace-agent.github.io>

Abstract

Large language model (LLM) applications such as agents and domain-specific reasoning increasingly rely on *context adaptation*—modifying inputs with instructions, strategies, or evidence, rather than weight updates. Prior approaches improve usability but often suffer from brevity bias, which drops domain insights for concise summaries, and from context collapse, where iterative rewriting erodes details over time. Building on the adaptive memory introduced by Dynamic Cheatsheet, we introduce ACE (Agentic Context Engineering), a framework that treats contexts as evolving playbooks that accumulate, refine, and organize strategies through a modular process of generation, reflection, and curation. ACE prevents collapse with structured, incremental updates that preserve detailed knowledge and scale with long-context models. Across agent and domain-specific benchmarks, ACE optimizes contexts both offline (*e.g.*, system prompts) and online (*e.g.*, agent memory), consistently outperforming strong baselines: +10.6% on agents and +8.6% on finance, while significantly reducing adaptation latency and rollout cost. Notably, ACE could adapt effectively without labeled supervision and instead by leveraging natural execution feedback. On the AppWorld leaderboard, ACE matches the top-ranked production-level agent on the overall average and surpasses it on the harder test-challenge split, despite using a smaller open-source model. These results show that comprehensive, evolving contexts enable scalable, efficient, and self-improving LLM systems with low overhead.

Context engineering as opposed to fine-tuning in the age of large models?

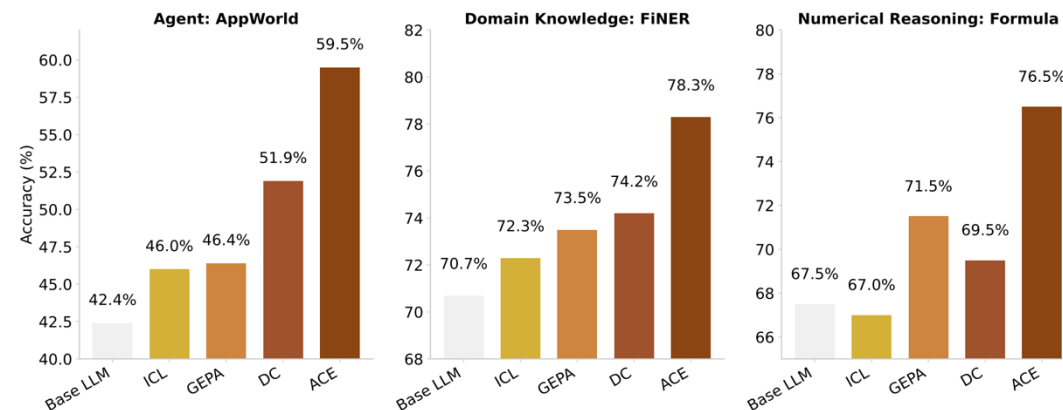


Figure 1: **Overall Performance Results.** Our proposed framework, ACE, consistently outperforms strong baselines across agent and domain-specific reasoning tasks.

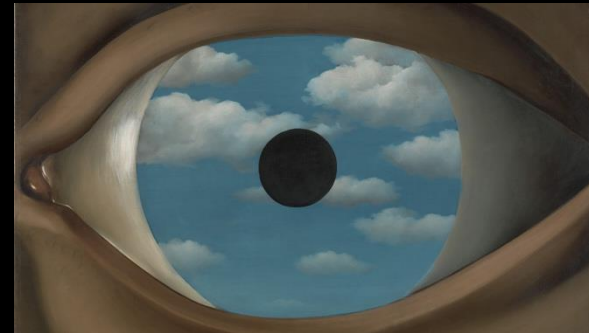
Visual Generation



“An IT-guy trying to fix hardware of a PC tower is being tangled by the PC cables like Laokoon. Marble, copy after Hellenistic original from ca. 200 BC. Found in the Baths of Trajan, 1506.”



“About 99% of the time, the right time is right now in René Magritte style.”



“The False Mirror” (1928)



“The Son of Man” (1964)

2023

2024

2025

2026

Video Generation (2023-2026)

<https://x.com/minchoi/status/2029409078775029860>

MLMI17:



Infinite Interactive Worlds

Project Genie. Google DeepMind. 2026.



Sora. OpenAI. 2024.
Demonstrated the power of Diffusion Transformer (DiT)



Sora: Video Generation Models as World Simulators. OpenAI. 2024.



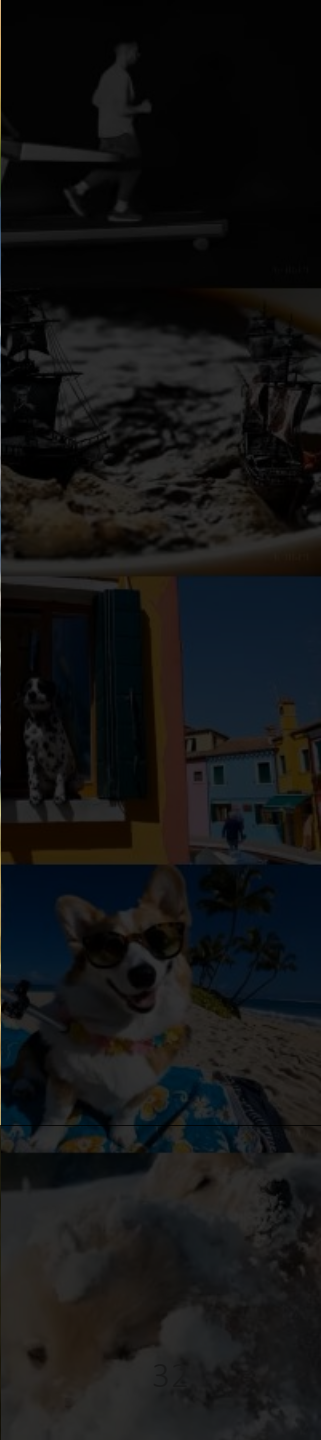
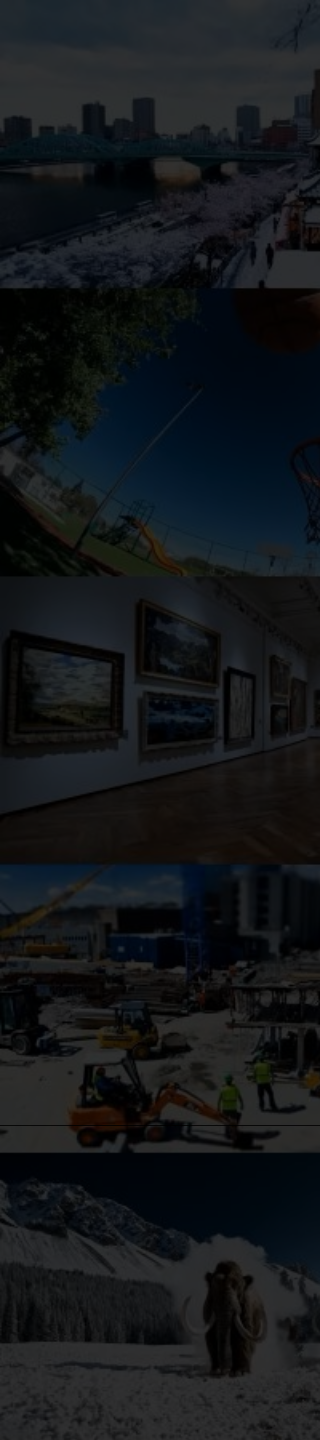
Pixels Defying Physics

Sora: Video Generation Models as World Simulators. OpenAI. 2024.



Pixels Defying Physics

Sora: Video Generation Models as World Simulators. OpenAI. 2024.





Reality Check

First frame from Physics IQ Benchmark, video generated by Veo 3.1 from Google DeepMind with the prompt: “what will happen if the spinning wheel keeps turning and the stick hits the first block on the right.” Adapted from VDAWorld (O'Mahony et al. 2025).



WorldScore

A Unified Evaluation Benchmark for World Generation

Models	WorldScore-Static	WorldScore-Dynamic	Camera Ctrl	Object Ctrl	Content Align	3D Consist	Photo Consist	Style Consist	Subjective Qual	Motion Acc	Motion Mag	Motion Smooth
Gen-3	60.71	<u>57.58</u>	29.47	62.92	50.49	68.31	87.09	62.82	63.85	54.53	27.48	68.87
Hailuo	57.55	56.36	22.39	69.56	<u>73.53</u>	67.18	62.82	54.91	52.44	63.46	27.20	70.07
DynamiCrafter	52.09	47.19	25.15	47.36	25.00	72.90	60.95	78.85	54.40	41.11	39.25	26.92
VideoCrafter1-T2V	47.10	43.54	21.61	50.44	60.78	64.86	51.36	38.05	42.63	11.76	75.00	18.87
VideoCrafter1-I2V	50.47	47.64	25.46	24.25	35.27	74.42	73.89	65.17	54.85	55.63	25.00	42.49
VideoCrafter2	52.57	47.49	28.92	39.07	72.46	65.14	61.85	43.79	56.74	47.12	30.40	29.39
T2V-Turbo	45.65	40.20	27.80	30.68	69.14	38.72	34.84	49.65	<u>68.74</u>	34.87	40.09	7.48
EasyAnimate	52.85	51.65	26.72	54.50	50.76	67.29	47.35	73.05	50.31	<u>75.00</u>	31.16	40.32
CogVideoX-T2V	54.18	48.79	40.22	51.05	68.12	68.81	64.20	42.19	44.67	25.00	<u>47.31</u>	36.28
CogVideoX-I2V	62.15	59.12	38.27	40.07	36.73	86.21	88.12	<u>83.22</u>	62.44	69.56	26.42	60.15
Allegro	55.31	51.97	24.84	57.47	51.48	70.50	69.89	65.60	47.41	54.39	40.28	37.81
Vchitect-2.0	42.28	38.47	26.55	49.54	65.75	41.53	42.30	25.69	44.58	33.59	33.81	21.31
SceneScape	50.73	35.51	84.99	47.44	28.64	76.54	62.88	21.85	32.75	0.00	0.00	0.00
Text2Room	62.10	43.47	94.01	38.93	50.79	<u>88.71</u>	88.36	37.23	36.69	0.00	0.00	0.00
LucidDreamer	70.40	49.28	88.93	41.18	75.00	90.37	90.20	48.10	58.99	0.00	0.00	0.00
WonderJourney	63.75	44.63	84.60	37.10	35.54	80.60	79.03	62.82	66.56	0.00	0.00	0.00
InvisibleStitch	61.12	42.78	<u>93.20</u>	36.51	29.53	88.51	<u>89.19</u>	32.37	58.50	0.00	0.00	0.00
WonderWorld	<u>72.69</u>	50.88	92.98	51.76	71.25	86.87	85.56	70.57	49.81	0.00	0.00	0.00
4D-fy	27.98	32.10	69.92	55.09	0.85	35.47	1.59	32.04	0.89	22.22	22.88	80.06
LTX-Video	55.44	56.54	25.06	53.41	39.73	78.41	88.92	53.50	49.08	76.22	29.95	<u>71.09</u>
Voyager	77.62	54.53	85.95	<u>66.92</u>	68.92	81.56	85.99	84.89	71.09	0.00	0.00	0.00
Wan2.1	57.56	52.85	23.53	40.32	45.44	78.74	78.36	77.18	59.38	54.27	33.26	38.05

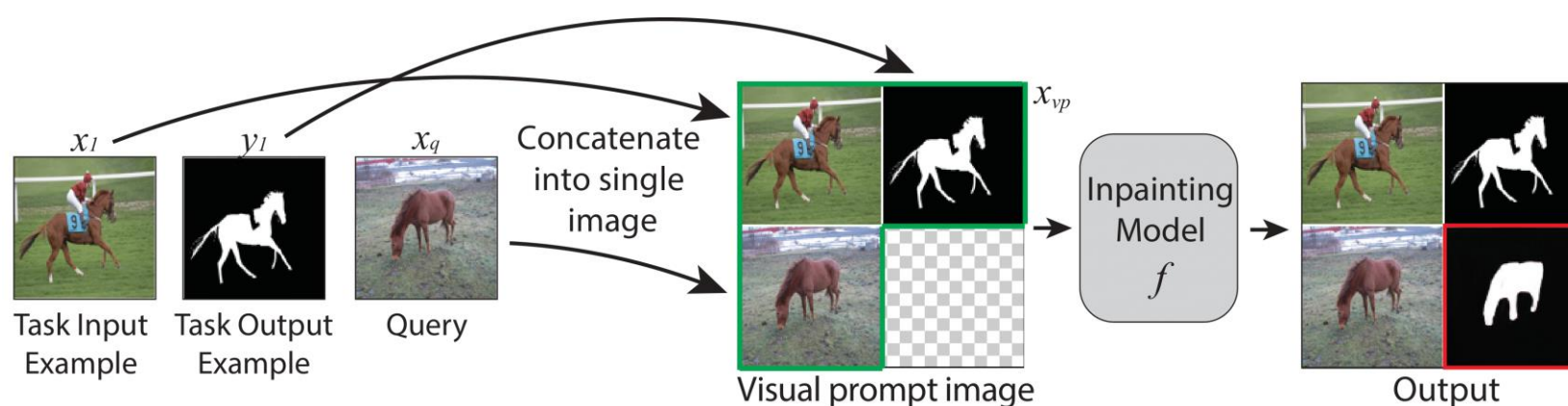
Visual Understanding via Conditional Generation

Visual Prompting via Image Inpainting

Amir Bar*^{1,2}, Yossi Gandelsman*¹, Trevor Darrell¹, Amir Globerson², Alexei A. Efros¹

UC Berkeley¹

Tel Aviv University²



Edge detection



Colorization



Inpainting



Segmentation



Style transfer

Visual Understanding via Conditional Generation

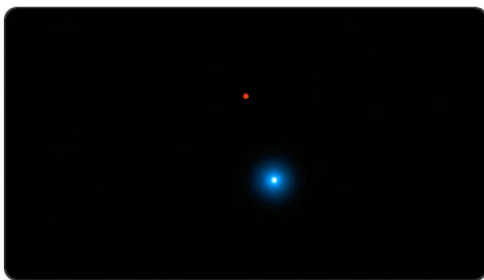
Video models are zero-shot learners and reasoners

Thaddäus Wiedemer*, Yuxuan Li, Paul Vicol, Shixiang Shane Gu, Nick Matarese, Kevin Swersky, Been Kim, Priyank Jaini*, Robert Geirhos*

Google DeepMind

* Joint leads

Perception



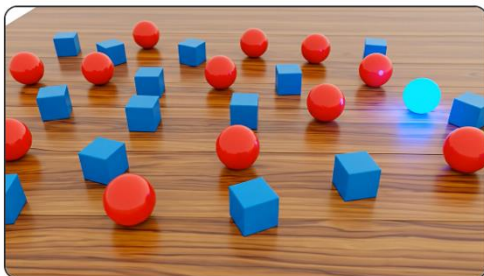
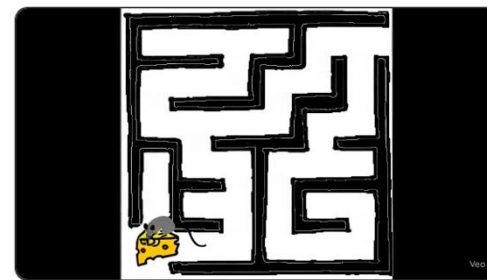
Modeling



Manipulation

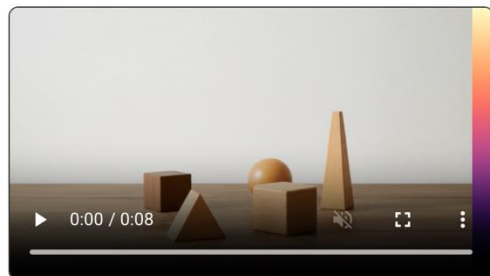


Reasoning



Visual Understanding via Conditional Generation

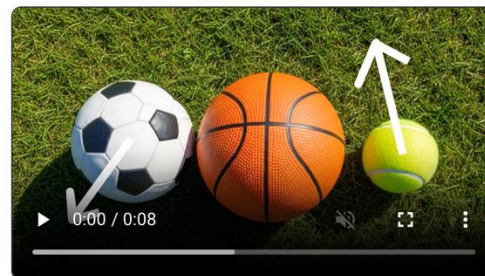
Failure cases (click to expand) ▼



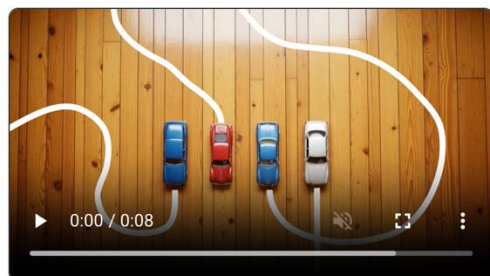
Monocular depth estimation



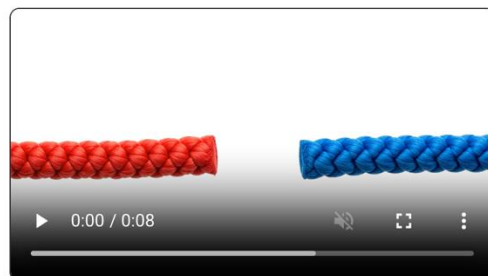
Monocular surface normal estimation



Force prompting



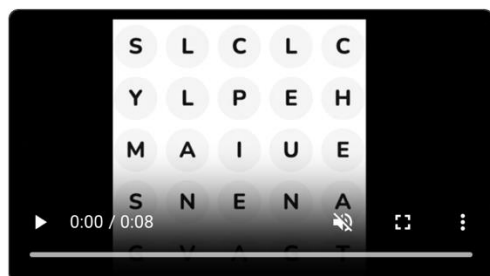
Motion trajectory prompting



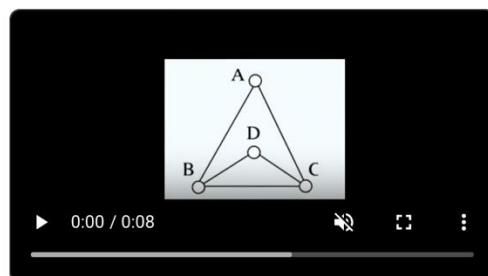
Tying the knot



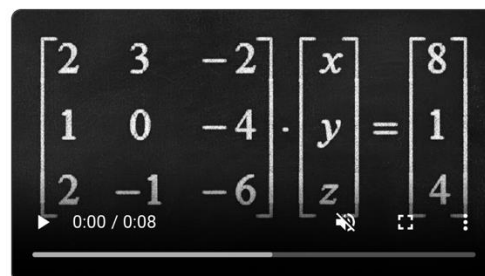
Connect the path puzzle



Letter word search



Eulerian path



Solving linear equations

Visual Understanding via Conditional Generation



automatic 3D data labelling pipeline using Gemini (Nano Banana 2)

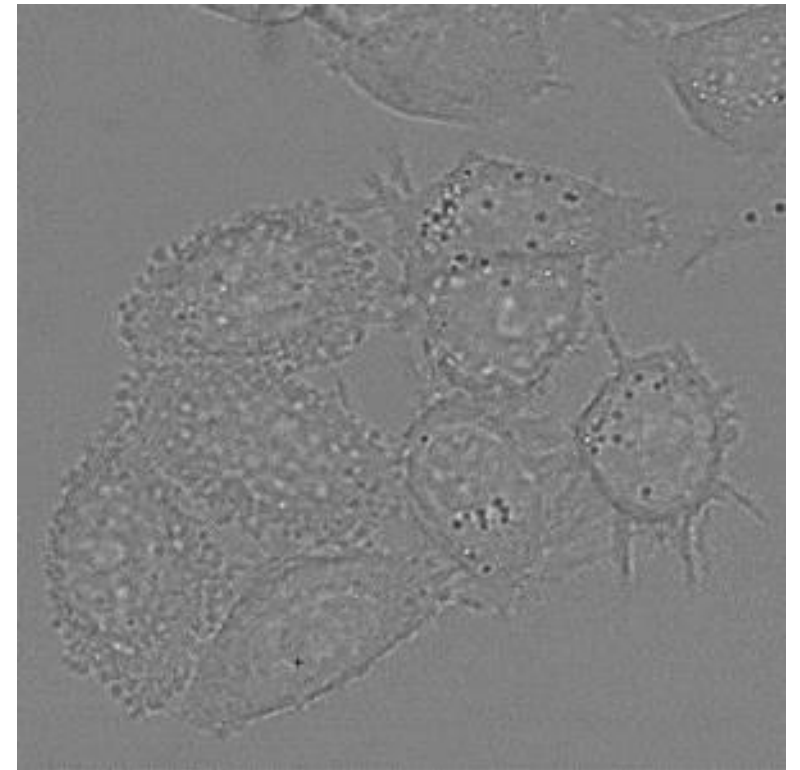
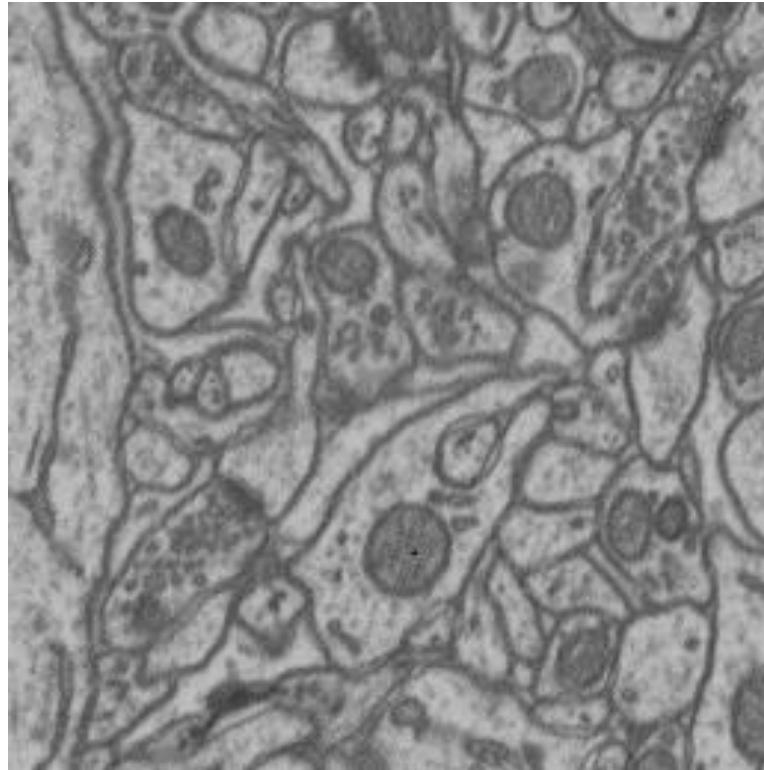
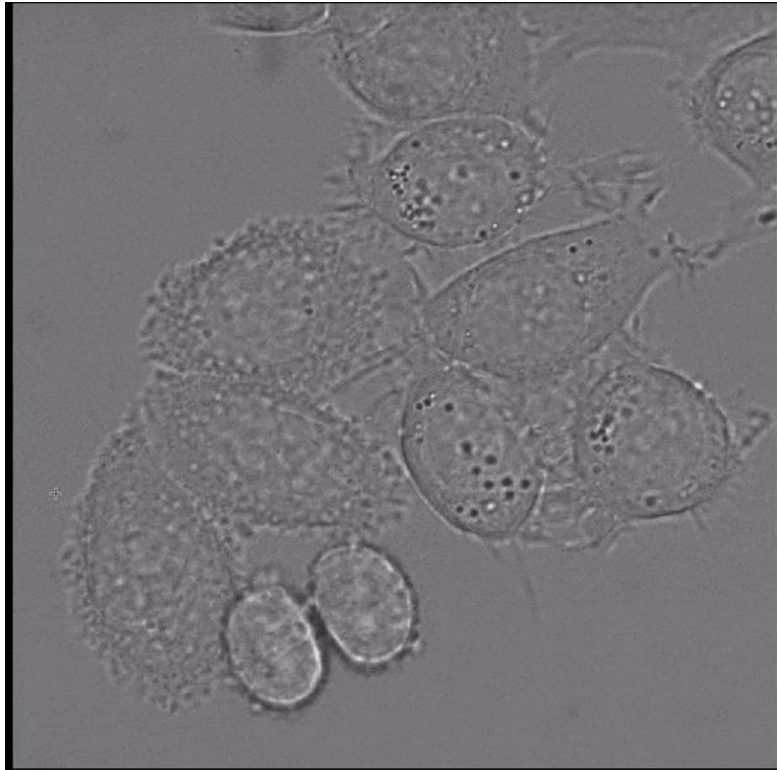
A photograph of three penguins on a sandy beach. The penguin on the left is facing away from the camera, looking towards the ocean. The two penguins on the right are facing the camera. A semi-transparent search bar is overlaid on the image, positioned over the two penguins on the right. The search bar contains a magnifying glass icon on the left and an 'X' icon on the right. The background shows gentle waves on the ocean under a clear sky.

Introducing Meta Segment Anything Model 3 (SAM 3)

SAM 3: Segment Anything with Concepts. Meta. 2025. <https://ai.meta.com/research/sam3/>

Segment Anything Model 3 (SAM 3)

Segment Anything for Microscopy



<https://github.com/computational-cell-analytics/micro-sam>

SAM 3D

GPT-4



SORA



**Everyone else scaling
to 10T parameters...**



**What remarkable instruments!
Time to learn to **conduct** them!**

Genie

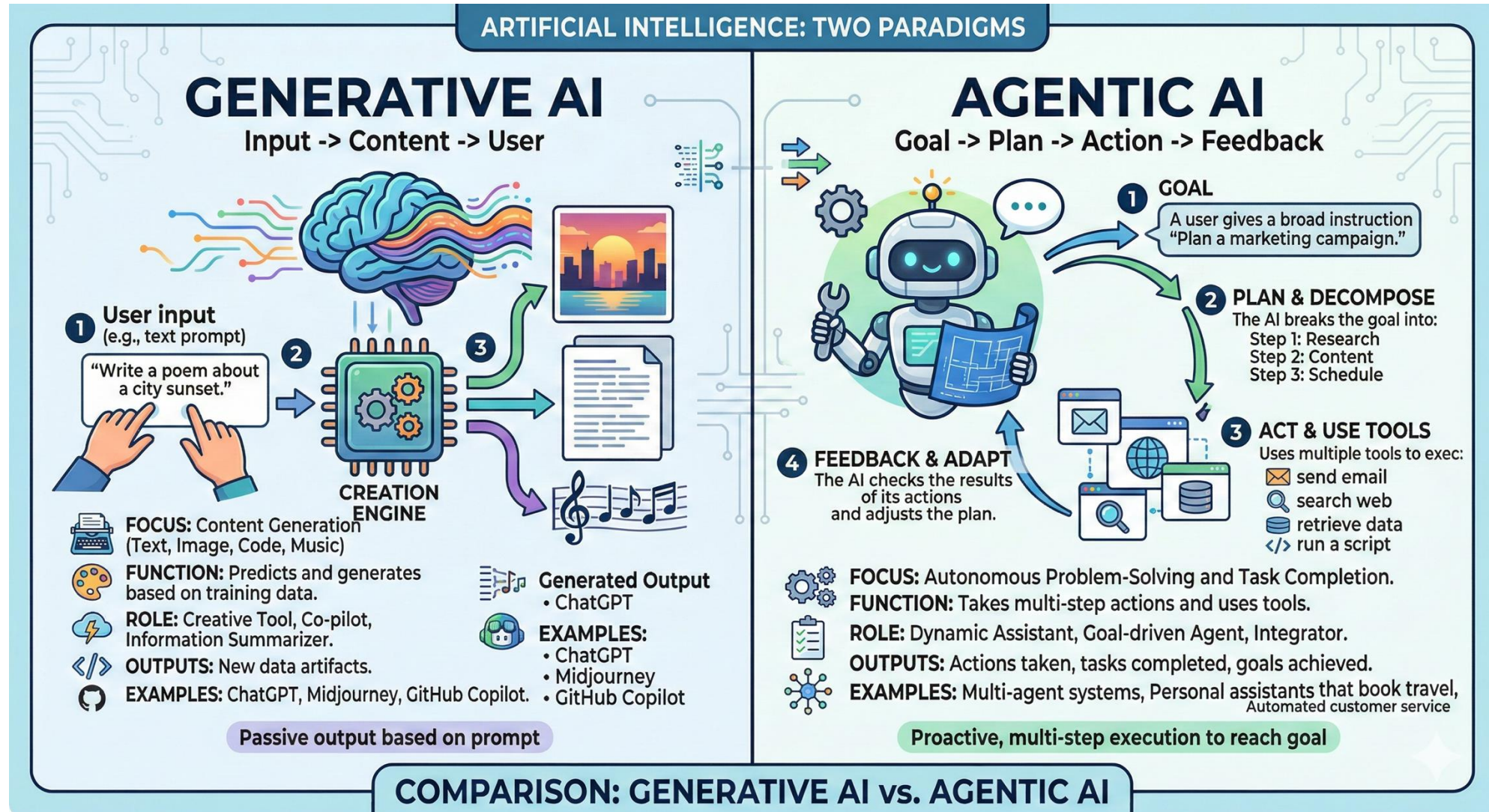


Me with Megalo-Modelo-Phobia.

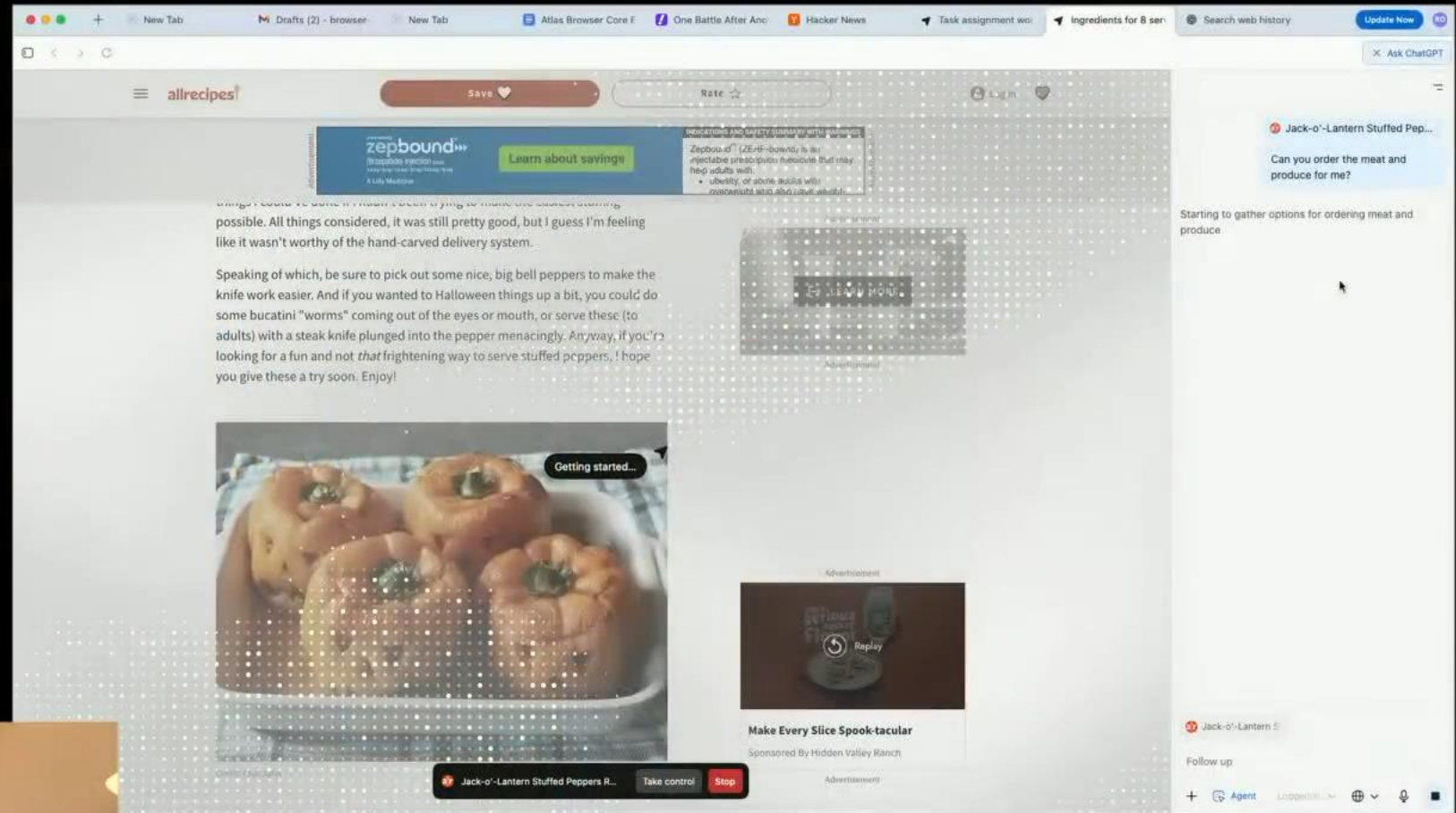


Part 2 – Expanding Horizons

Agentic AI Systems (2024-now)



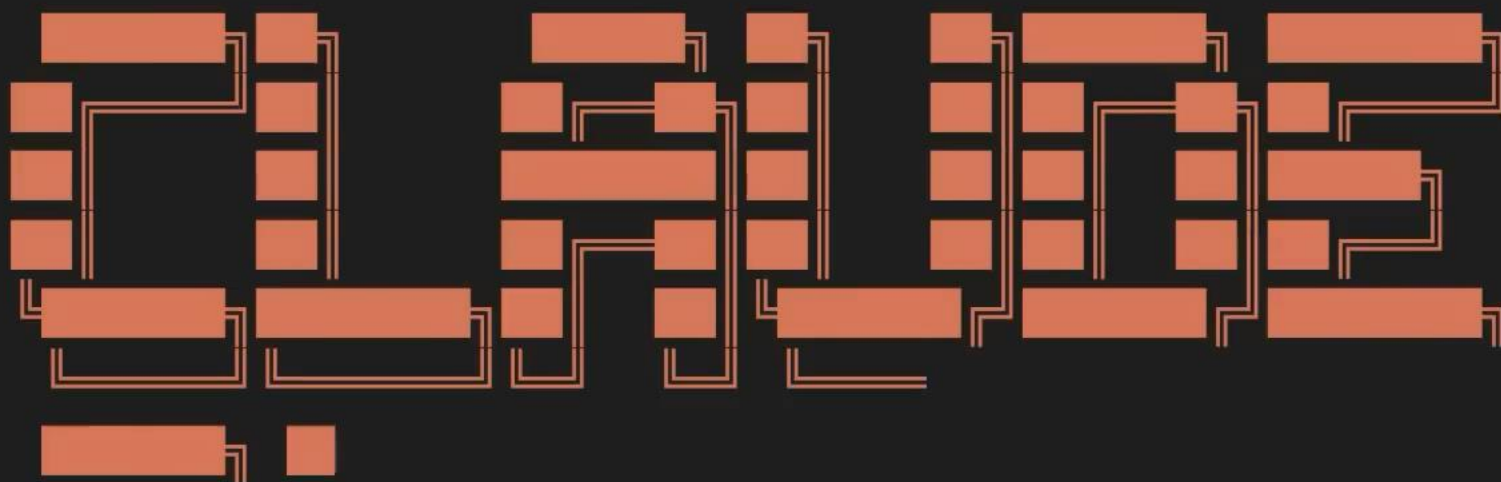
ChatGPT Atlas



Codex

Introducing the Codex app. OpenAI. 2026. <https://www.youtube.com/watch?v=HFM3se4INiw>

* Welcome to the **Claude Code** research preview!



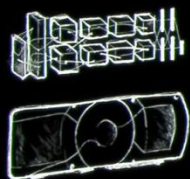
Claude Code

Cursor



What's Next?

2012 ALEXNET



PERCEPTION AI

SPEECH RECOGNITION
DEEP RECSYS
MEDICAL IMAGING

GENERATIVE AI

DIGITAL MARKETING
CONTENT CREATION

AGENTIC AI

CODING ASSISTANT
CUSTOMER SERVICE
PATIENT CARE

PHYSICAL AI

SELF-DRIVING CARS
GENERAL ROBOTICS

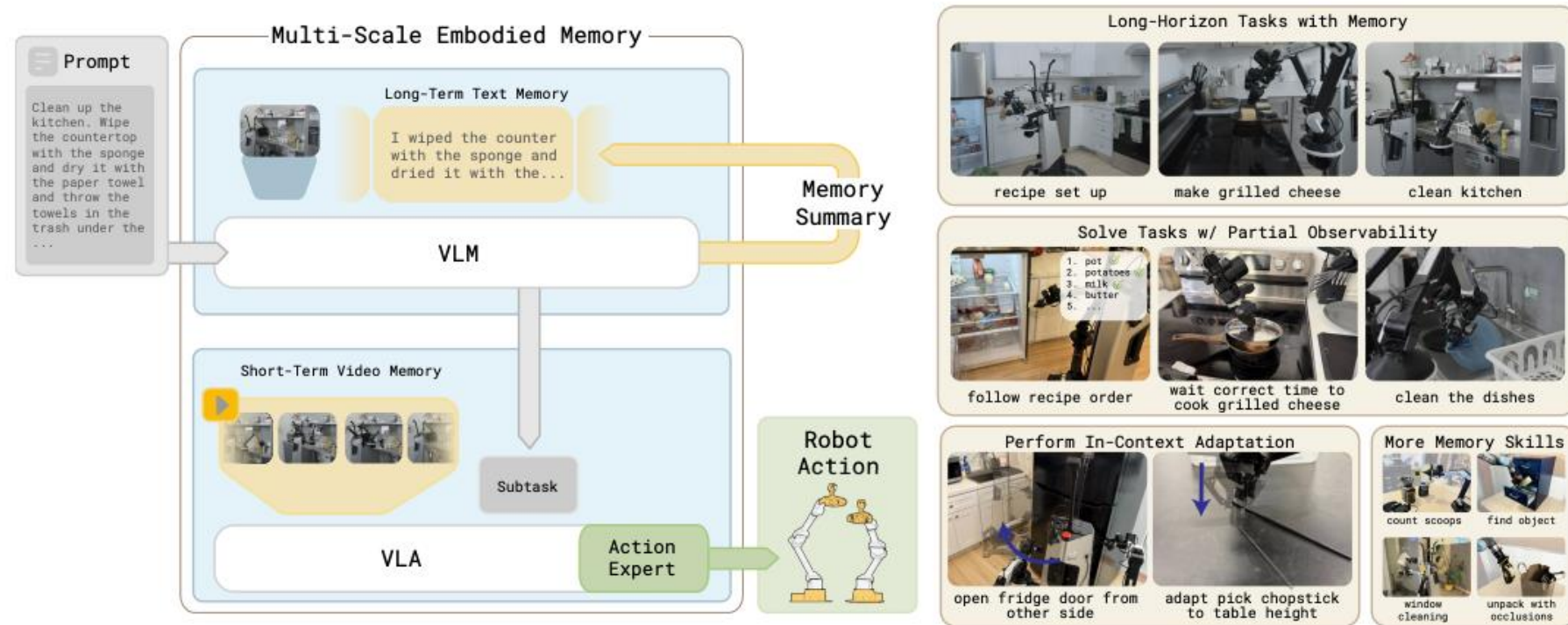
Today's Roadblocks to Physical AI

- Data collection cost
- Physical accuracy (hallucination)
- Memory for long-horizon tasks
- Hardware dexterity
- Latency in real-time deployment
- ...

MEM: Multi-Scale Embodied Memory for Vision Language Action Models

Marcel Torne^{*12†} Karl Pertsch^{*1} Homer Walke^{3†} Kyle Vedder¹ Suraj Nair¹ Brian Ichter¹
Allen Z. Ren¹ Haohuan Wang¹ Jiaming Tang^{14†} Kyle Stachowicz^{13†} Karan Dhabalia¹ Michael Equi¹
Quan Vuong¹ Jost Tobias Springenberg¹ Sergey Levine¹ Chelsea Finn¹ Danny Driess¹

<https://pi.website/research/memory>



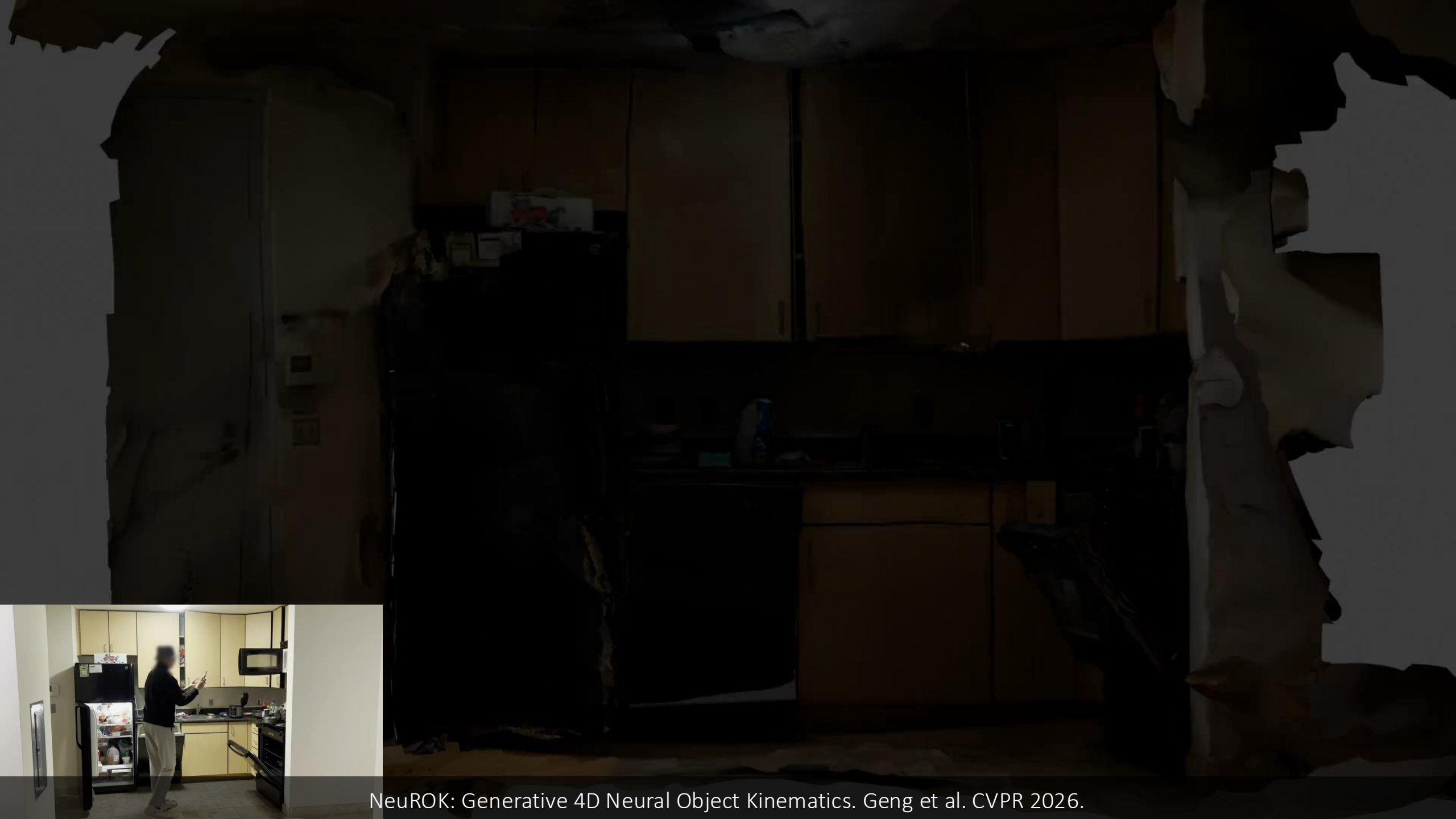
Generative Simulation







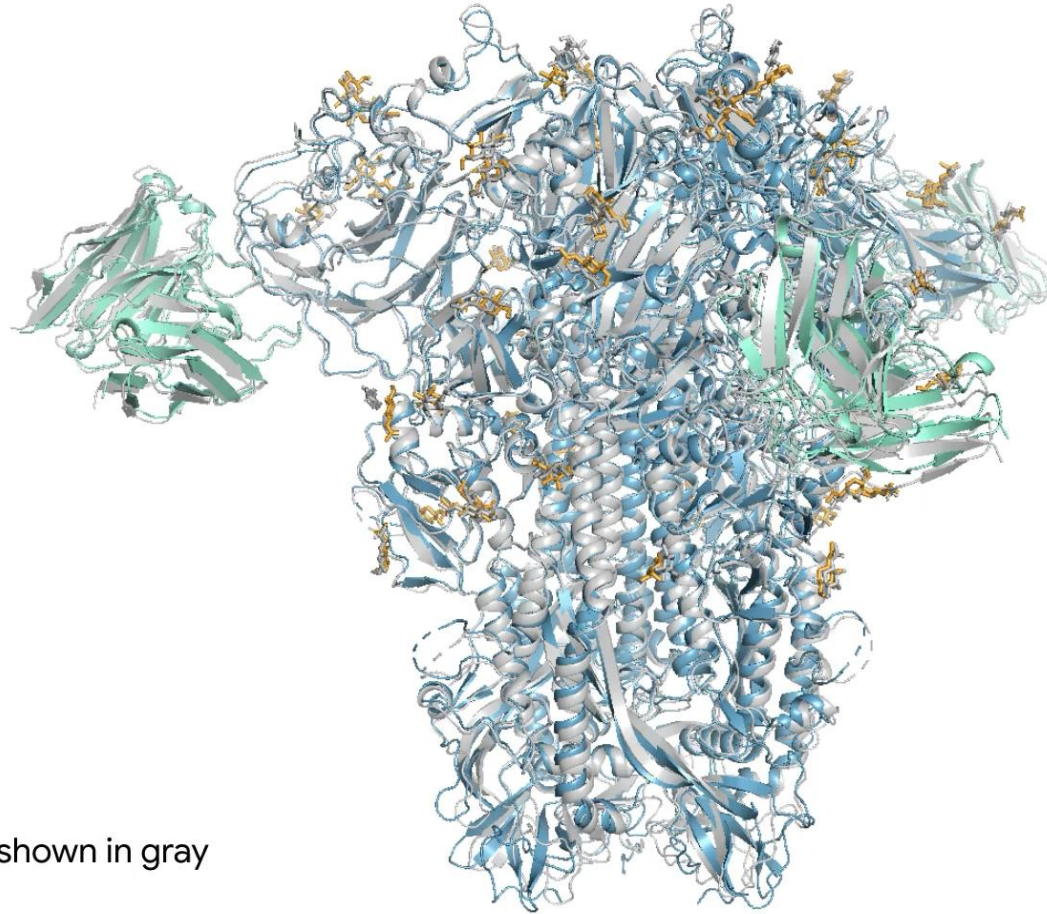
NeuROK: Generative 4D Neural Object Kinematics. Geng et al. CVPR 2026.



NeuROK: Generative 4D Neural Object Kinematics. Geng et al. CVPR 2026.

AI for Science

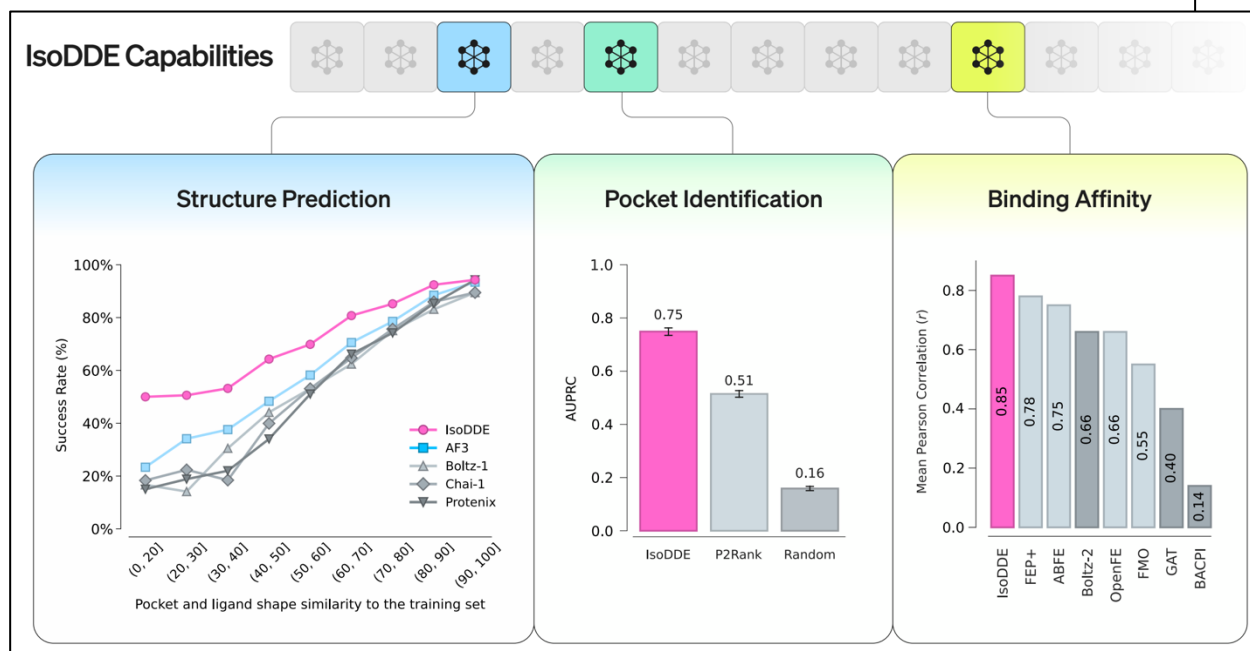
7PNM



Ground truth shown in gray

AlphaFold 3 – diffusion model predicting 3D folding structure of proteins

AI for Science

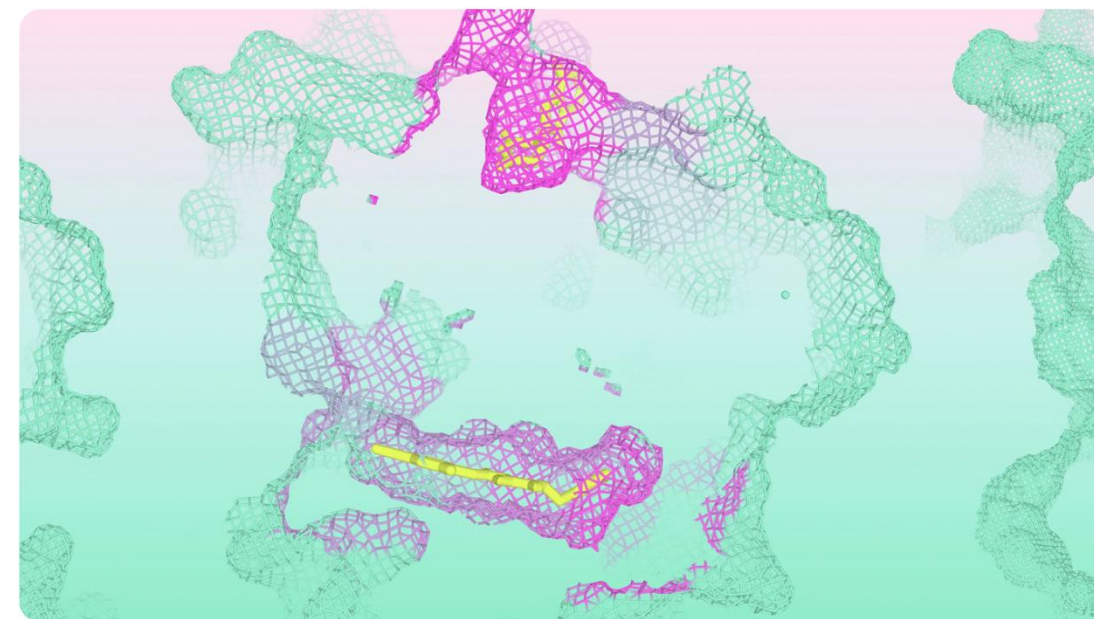


The Isomorphic Labs Drug Design Engine unlocks a new frontier beyond AlphaFold

FEBRUARY 10, 2026

5 MIN READ

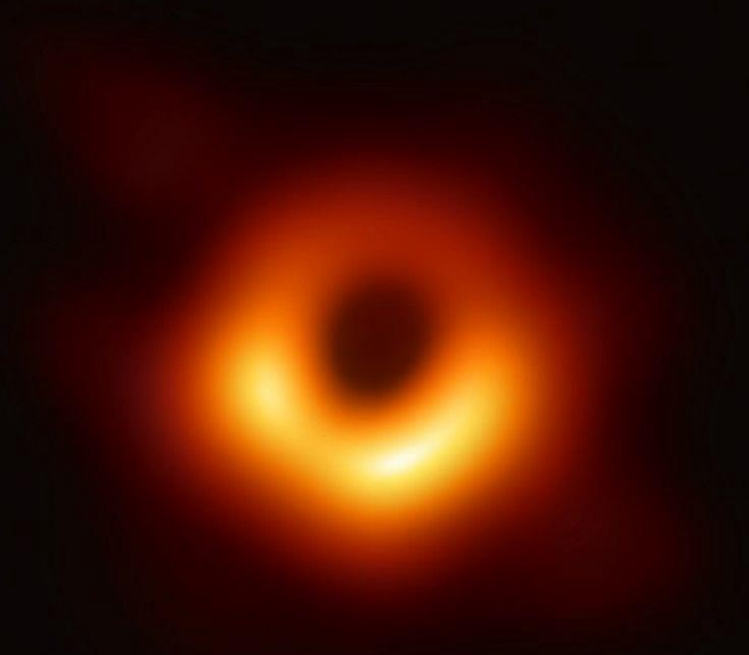
COPY URL



- Structure Prediction of Truly Novel Systems
- Opening a New Window for Complex Biologics
- A New Gold-Standard for Binding Affinity Prediction
- Expanding the Ligandable Proteome
- Advancing Drug Discovery

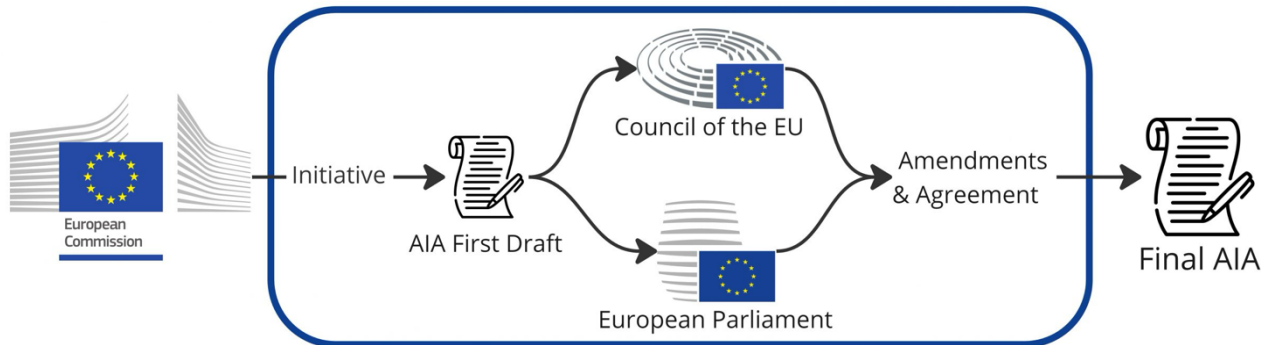
Today, we are excited to share an update on our progress towards a new frontier of drug design. We have unlocked a new paradigm of predictive accuracy in understanding our biomolecular world, allowing us to rationally design new medicines on a computer with unprecedented understanding and precision.

First Image of a Black Hole



reconstructed from massive amounts of radio signals using Interferometry
Multi-view Geometry in Fourier domain (like MRI scans)

Ethics and Governance



Ordinary Legislative Procedure

<https://artificialintelligenceact.eu/>



Visual General Intelligence

-Vision Research Toward the AGI Era-

CVPR 2026 Workshop

Invited Speakers

Robert Geirhos
Google DeepMind



[View Profile](#)

Aditi Raghunathan
Carnegie Mellon University



[View Profile](#)

Matt Deitke
Meta Superintelligence Lab



[View Profile](#)

Kristen Grauman
University of Texas, Austin / Meta



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Yuki M. Asano
University of Technology Nuremberg



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Alexei A. Efros
University of California, Berkeley



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Jamie Shotton
Wayve



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Kaiming He
Massachusetts Institute of Technology / Google DeepMind



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Andrea Vedaldi
University of Oxford / Meta



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Summary – Applications & Challenges

- The Era of Large Models
 - Multimodal Language Models (MLLMs)
 - In-context learning
 - Image and video generation, interactive world generation
 - Segment Anything (SAM), SAM 3D
- Expanding Horizons
 - Agentic AI
 - Physical AI
 - AI for Science
 - Ethics and governance

Topics We Didn't Have Time to Cover

- Latent predictive world models: [JEPA](#), [V-JEPA](#)
- Understanding x generation: [Representation Autoencoders \(RAEs\)](#)
- Vision as a first-class citizen: [Multimodal Pretraining](#)
- Spatial reasoning: [Theory of Space](#), [Spatial Mental Modeling](#)
- Autonomous driving: [UniAD](#), [Tesla FSD](#), [Waymo](#), [Wayve](#)
- Vision-Language-Action (VLA) models: [RT-X](#), [\$\pi_0\$](#) , [Gemini Robotics](#)
- ...

Codex, Why You Are So Slow?

An AI-generated song for vibe coders
by Johnny Vibe Building

- Grok for fast & smooth video motion
- Midjourney for nice Niji 7 anime style
- Kimi for aesthetic prompts
- Suno v5: Amazing!
- OpenAI: Please make GPT 5.3 xhigh lightning fast! 🤔

<https://www.youtube.com/watch?v=bSzGfPLgsg>

MLM17: Advanced Computer Vision

